



Chapter 4: The Final Code

Python

Students discover the origins of the Darkness and the key to defeating it once and for all!

Students use compound conditionals, functions, data analysis, & different objects.

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#5. Over Abyss

Level Overview and Solutions

Over Abyss

Intro

Mouse is on the other side but those platforms look like they're falling!

Default Code

```
1 mouse.moveUp(5)
2
3 while hero.getDistanceTo(mouse) > 1:
4     if hero.canMoveRight():
5         hero.moveRight()
6     elif hero.canMoveUp():
7         hero.moveUp()
8     elif hero.canMoveDown():
9         hero.moveDown()
10    elif hero.canMoveLeft():
11        hero.moveLeft()
```

Over Abyss Solution

```
1 mouse.moveUp(5)
2
3 while hero.getDistanceTo(mouse) > 1:
4     if hero.canMoveRight():
5         hero.moveRight()
6     elif hero.canMoveUp():
7         hero.moveUp()
8     elif hero.canMoveDown():
9         hero.moveDown()
10    elif hero.canMoveLeft():
11        hero.moveLeft()
```

#9. What Else

Level Overview and Solutions

What Else

Intro

Mouse is on the other side again and ugh, there's more of those shaky platforms!

Default Code

```
1 mouse.moveRight(7)
2
3 while hero.getDistanceTo(mouse) > 1:
4     if hero.canMoveUp():
5         hero.moveUp()
6     elif hero.canMoveRight():
7         hero.moveRight()
```

What Else Solution

```
1 mouse.moveRight(7)
2
3 while hero.getDistanceTo(mouse) > 1:
4     if hero.canMoveUp():
5         hero.moveUp()
6     elif hero.canMoveRight():
7         hero.moveRight()
8     elif hero.canMoveDown():
9         hero.moveDown()
10    elif hero.canMoveLeft():
11        hero.moveLeft()
```

#10. Run of Faith

Level Overview and Solutions

Run of Faith

Intro

How does Mouse keep getting across?? There are a lot of platforms here so step carefully!

Default Code

```
1 mouse.digUp()
2 mouse.moveUp(1)
3
4 mouse.digUp()
5 mouse.moveUp(1)
6
7 while hero.getDistanceTo(mouse) > 1:
8     if hero.canMoveRight():
9         hero.moveRight()
```

Run of Faith Solution

```
1 mouse.digUp()
2 mouse.moveUp(1)
3
4 mouse.digUp()
5 mouse.moveUp(1)
6
7 while hero.getDistanceTo(mouse) > 1:
8     if hero.canMoveRight():
9         hero.moveRight()
10    elif hero.canMoveUp():
11        hero.moveUp()
12    elif hero.canMoveDown():
13        hero.moveDown()
14    elif hero.canMoveLeft():
15        hero.moveLeft()
```

#11. Platform Run

Level Overview and Solutions

Platform Run

Intro

Ok. I really hope this is the last set of platforms. I'm losing steam!

Default Code

1

Platform Run Solution

```
1 mouse.digRight()
2 mouse.moveRight(2)
3 mouse.digRight()
4 mouse.moveRight(2)
5
6 while hero.getDistanceTo(mouse) > 1:
7     if hero.canMoveRight():
8         hero.moveRight()
9     elif hero.canMoveUp():
10        hero.moveUp()
11    elif hero.canMoveDown():
12        hero.moveDown()
13    elif hero.canMoveLeft():
14        hero.moveLeft()
```

#14. Less Than Close

Level Overview and Solutions

Less Than Close

Intro

Those Splinters have been destroying our lands over and over again! With my code and your Aodus, I think we can get rid of those Splinters for good.

Default Code

```
1 defeatedEnemies = 0
2
3 enemy = hero.findNearestEnemy()
4 while defeatedEnemies < 3:
5     if hero.getDistanceTo(enemy) <= 2:
6         hero.castEarthWave()
7         defeatedEnemies += 1
8         hero.moveRight(2)
9         enemy = hero.findNearestEnemy()
```

Less Than Close Solution

```
1 defeatedEnemies = 0
2
3 enemy = hero.findNearestEnemy();
4 while defeatedEnemies < 3:
5     if hero.getDistanceTo(enemy) <= 2:
6         hero.castEarthWave()
7         defeatedEnemies += 1
8         hero.moveRight(2)
9         enemy = hero.findNearestEnemy();
10
11 hero.moveDown()
```

#18. Close Line

Level Overview and Solutions

Close Line

Intro

More Splinters! They just keep coming.

Default Code

```
1 hero.moveRight(1)
2 defeatedEnemies = 0
3
4 enemy = hero.findNearestEnemy()
5 while defeatedEnemies < 3:
6     enemyDistance = hero.getDistanceTo(enemy)|
```

Close Line Solution

```
1 hero.moveRight(1)
2 defeatedEnemies = 0
3
4 enemy = hero.findNearestEnemy()
5 while defeatedEnemies < 3:
6     if hero.getDistanceTo(enemy) <= 2:
7         hero.castEarthWave()
8         defeatedEnemies += 1
9         hero.moveUp(2)
10        enemy = hero.findNearestEnemy()
11
12 hero.jumpRight()
13 hero.moveRight(3)|
```

#21. Between Enemies

Level Overview and Solutions

Between Enemies

Intro

Time to try out the `getDistanceBetween` spell. It'll help you defeat the Splinters while keeping my people safe!

Default Code

```
1 defeatedEnemies = 0
2
3 enemy = hero.findNearestEnemy()
4 friend = hero.findNearestFriend()
5
6 while defeatedEnemies < 2:
7     if hero.getDistanceBetween(enemy, friend) >= 3:
8         hero.castEarthPit(enemy)
```

Between Enemies Solution

```
1 defeatedEnemies = 0
2
3 enemy = hero.findNearestEnemy()
4 friend = hero.findNearestFriend()
5
6 while defeatedEnemies < 2:
7     if hero.getDistanceBetween(enemy, friend) >= 3:
8         hero.castEarthPit(enemy)
9         defeatedEnemies += 1
10        hero.moveUp(2)
11        enemy = hero.findNearestEnemy()
12        friend = hero.findNearestFriend()
```

#22. Splinters

Level Overview and Solutions

Splinters

Intro

Splinters, Splinters, everywhere! When will they stop destroying our lands?

Default Code

```
1 hero.moveRight(2)
2
3 defeatedEnemies = 0
4
5 while defeatedEnemies < 2:
6
7     enemy = hero.findNearestEnemy()
8     friend = hero.findNearestFriend()
9
10    distanceBetween = hero.getDistanceBetween(enemy, friend)
```

Splinters Solution

```
1 hero.moveRight(2)
2
3 defeatedEnemies = 0
4
5 while defeatedEnemies < 2:
6
7     enemy = hero.findNearestEnemy()
8     friend = hero.findNearestFriend()
9
10    distanceBetween = hero.getDistanceBetween(enemy, friend)
11    if distanceBetween >= 3:
12        hero.castEarthPit(enemy)
13        defeatedEnemies += 1
14        hero.jumpRight()
15        hero.moveRight(1)
```

#25. What If If

Level Overview and Solutions

What If If

Intro

Hmmmm. There are a lot more Tengshe here so you'll need to be extra careful.

Default Code

```
1 defeatedEnemies = 0
2
3 friend = hero.findNearestFriend()
4 enemy = hero.findNearestEnemy()
5
6 while defeatedEnemies < 3:
7     distanceFriend = hero.getDistanceTo(friend)
8     distanceEnemy = hero.getDistanceTo(enemy)
9     if distanceFriend > 2:
10         if distanceEnemy <= 2:
11             hero.castEarthWave()
```

What If If Solution

```
1 defeatedEnemies = 0
2
3 friend = hero.findNearestFriend()
4 enemy = hero.findNearestEnemy()
5
6
7 while defeatedEnemies < 3:
8     distanceFriend = hero.getDistanceTo(friend)
9     distanceEnemy = hero.getDistanceTo(enemy)
10    if distanceFriend > 2:
11        if distanceEnemy <= 2:
12            hero.castEarthWave()
13            defeatedEnemies += 1
14            hero.sneakRight(2)
15            friend = hero.findNearestFriend()
16            enemy = hero.findNearestEnemy()
```


#28. What And

Level Overview and Solutions

What And

Intro

Another crowded place! At least you can **combine conditions** to simplify the code.

Default Code

```
1 defeatedEnemies = 0
2
3 hero.sneakRight(1)
4 friend = hero.findNearestFriend()
5 enemy = hero.findNearestEnemy()
6
7 while defeatedEnemies < 3:
8     distanceFriend = hero.getDistanceTo(friend)
9     distanceEnemy = hero.getDistanceTo(enemy)
10    if distanceFriend > 2 and distanceEnemy <= 2:
11        hero.castEarthWave()
```

What And Solution

```
1 defeatedEnemies = 0
2
3 hero.sneakRight(1)
4 friend = hero.findNearestFriend()
5 enemy = hero.findNearestEnemy()
6
7
8 while defeatedEnemies < 3:
9     distanceFriend = hero.getDistanceTo(friend)
10    distanceEnemy = hero.getDistanceTo(enemy)
11    if distanceFriend > 2 and distanceEnemy <= 2:
12        hero.castEarthWave()
13        defeatedEnemies += 1
14        hero.sneakRight(2)
15        enemy = hero.findNearestEnemy()
```

#29. And Three More Times

Level Overview and Solutions

And Three More Times

Intro

Time to try out compound conditionals on your own! See if you can defeat these Splinters.

Default Code

```
1 defeatedEnemies = 0
2
3 friend = hero.findNearestFriend()
4 enemy = hero.findNearestEnemy()
5
6 while defeatedEnemies < 3:
7     distanceFriend = hero.getDistanceTo(friend)
8     distanceEnemy = hero.getDistanceTo(enemy)
```

And Three More Times Solution

```
1 defeatedEnemies = 0
2
3 friend = hero.findNearestFriend()
4 enemy = hero.findNearestEnemy()
5
6 while defeatedEnemies < 3:
7     distanceFriend = hero.getDistanceTo(friend)
8     distanceEnemy = hero.getDistanceTo(enemy)
9     if distanceFriend > 2 and distanceEnemy <= 2:
10         hero.castEarthWave()
11         defeatedEnemies += 1
12         hero.sneakUp(2)
13         friend = hero.findNearestFriend()
14         enemy = hero.findNearestEnemy()
```

#32. If Or If

Level Overview and Solutions

If Or If

Intro

We all need to get across, but there are now both **Shadow Seeds** and **Splinters** in the way!

Default Code

```
1 for i in range(7):
2     enemy = hero.findNearestEnemy()
3     distanceEnemy = hero.getDistanceTo(enemy)
4     if hero.canMoveRight() == False:
5         hero.castEarthWave()
```

If Or If Solution

```
1 for i in range(7):
2     enemy = hero.findNearestEnemy()
3     distanceEnemy = hero.getDistanceTo(enemy)
4     if hero.canMoveRight() == False:
5         hero.castEarthWave()
6     if distanceEnemy <= 2:
7         hero.castEarthWave()
8     hero.moveRight(1)
```

#37. Just Rock It

Level Overview and Solutions

Just Rock It

Intro

You need to clear these paths, but now you can use **or** to simplify the code!

Default Code

```
1 for i in range(7):
2     enemy = hero.findNearestEnemy()
3     distanceEnemy = hero.getDistanceTo(enemy)
4     if distanceEnemy <= 2 or not hero.canMoveRight():
5         hero.castEarthWave()
```

Just Rock It Solution

```
1 for i in range(7):
2     enemy = hero.findNearestEnemy()
3     distanceEnemy = hero.getDistanceTo(enemy)
4     if distanceEnemy <= 2 or not hero.canMoveRight():
5         hero.castEarthWave()
6     hero.moveRight(1)
```

#38. Rocket Up

Level Overview and Solutions

Rocket Up

Intro

More Shadow Seeds and more Splinters. You know what to do! Think about what you need to check for your **conditional**.

Default Code

```
1 for i in range(6):  
2     enemy = hero.findNearestEnemy()
```

Rocket Up Solution

```
1 for i in range(6):  
2     enemy = hero.findNearestEnemy()  
3     distanceEnemy = hero.getDistanceTo(enemy)  
4     if not hero.canMoveUp() or distanceEnemy <= 2:  
5         hero.castEarthWave()  
6     hero.moveUp(1)
```

#39. Unblocker

Level Overview and Solutions

Unblocker

Intro

How are we going to get past all these **bridges**? Oh look, **Snikrep's villagers** are ready to help!

Default Code

```
1 for i in range(4):
2     pathUpFree = hero.canMoveUp()
3     pathDownFree = hero.canMoveDown()
```

Unblocker Solution

```
1 for i in range(4):
2     pathUpFree = hero.canMoveUp()
3     pathDownFree = hero.canMoveDown()
4
5     if not pathUpFree or not pathDownFree:
6         hero.castEarthWave()
7     hero.moveRight(2)
```

#40. Gardening

Level Overview and Solutions

Gardening

Intro

I think the **Star Well** is in that building up ahead. We just need to cross these bridges to get to it!

Default Code

1

Gardening Solution

```
1 for i in range(4):
2     pathUpFree = hero.canMoveUp()
3     pathDownFree = hero.canMoveDown()
4
5     if not pathUpFree or not pathDownFree:
6         hero.castEarthWave()
7     hero.moveRight(2)
8
9 hero.moveDown(2)
```

#43. Return to Haven

Level Overview and Solutions

Return to Haven

Intro

The **Time Totem** is in the town of Haven. The only way there is through the Phoenix Lands, but now it's full of Splinters and enemies!

Default Code

```
1 mouse.digUp()
2 mouse.moveRight(3)
3 mouse.moveUp()
4
5 for i in range(3):
6     hero.sneakRight(2)
```

Return to Haven Solution

```
1 mouse.digUp()
2 mouse.moveRight(3)
3 mouse.moveUp()
4
5 for i in range(3):
6     hero.sneakRight(2)
7     hero.sneakUp(2)
8
9 hero.jumpRight()
10 hero.moveDown(2)
```

#44. The Path Back

Level Overview and Solutions

The Path Back

Intro

I can see **Haven** up ahead past these ruins. Just a little farther!

Default Code

```
1 for i in range(3):
2     enemy = hero.findNearestEnemy()
3     hero.castIllusionMist(enemy)
4     hero.sneakLeft(3)
5
6 hero.moveUp(3)
```

The Path Back Solution

```
1 for i in range(3):
2     enemy = hero.findNearestEnemy()
3     hero.castIllusionMist(enemy)
4     hero.sneakLeft(3)
5
6 hero.moveUp(3)
7
8 for i in range(4):
9     hero.moveRight(2)
10    mouse.moveRight(2)
11
12 hero.moveRight()
```

#45. Hide and Sneak

Level Overview and Solutions

Hide and Sneak

Intro

We're finally in **Haven** but the streets are also full of Splinters! We'll need to **hide and sneak** past them to get to **Time Totem** at the center of the town.

Default Code

```
1 hero.hide()
2 enemy = hero.findNearestEnemy()
3
4 for i in range(4):
5     mouse.digRight()
```

Hide and Sneak Solution

```
1 hero.hide()
2 enemy = hero.findNearestEnemy()
3
4 for i in range(4):
5     mouse.digRight()
6
7
8 while hero.getDistanceTo(enemy) < 3:
9     hero.hide()
10
11 hero.moveRight(2)
12 hero.moveDown()
13
14 for i in range(2):
15     hero.jumpRight()
16     hero.jumpDown()
17
18 hero.jumpRight()
```

#48. Caterpillars

Level Overview and Solutions

Caterpillars

Intro

To make the Acodus, we need a special ingredient called **glow fruits**. They're only found in the **Spirit Lands**. That's where my trusty bunny comes in.

Default Code

```
1 noodles.moveUp(2)
2 noodles.moveRight(1)
3 noodles.moveDown(2)
4 noodles.moveRight(1)
```

Caterpillars Solution

```
1 noodles.moveUp(2)
2 noodles.moveRight(1)
3 noodles.moveDown(2)
4 noodles.moveRight(1)
5
6 noodles.moveUp(3)
7
8 noodles.moveUp(2)
9 noodles.moveRight(1)
10 noodles.moveDown(2)
11 noodles.moveRight(1)
12
13 noodles.moveDown(2)
14 noodles.moveRight(1)
15
16 noodles.moveUp(2)
17 noodles.moveRight(1)
18 noodles.moveDown(2)
19 noodles.moveRight(1)
```

#51. Pi Jump

Level Overview and Solutions

Pi Jump

Intro

This area of the Spirit Lands doesn't seem to have many glow fruits, but it's good to check.

Default Code

```
1 def jumpAround():
2     noodles.jumpUp(1)
3     noodles.jumpRight(1)
4     noodles.jumpDown(1)
5
6
7 jumpAround()
8 noodles.moveRight()
```

Pi Jump Solution

```
1 def jumpAround():
2     noodles.jumpUp(1)
3     noodles.jumpRight(1)
4     noodles.jumpDown(1)
5
6 jumpAround()
7 noodles.moveRight(1)
8 noodles.jumpUp(1)
9
10 jumpAround()
11 noodles.jumpDown()
12 noodles.moveRight(1)
13
14 jumpAround()
```

#54. Crystal Snake

Level Overview and Solutions

Crystal Snake

Intro

The path looks more complicated here. You'll need a different **function** to help Noodles continue her search.

Default Code

```
1 def snake():
2     noodles.moveUp(1)
3     noodles.moveRight(1)
4     noodles.moveDown(1)
5     noodles.moveRight(1)
6     # add an action here to finish the function
7
8
9
10 noodles.moveRight(1)
11 snake()
```

Crystal Snake Solution

```
1 def snake():
2     noodles.moveUp(1)
3     noodles.moveRight(1)
4     noodles.moveDown(1)
5     noodles.moveRight(1)
6     # add an action here to finish the function
7     noodles.moveUp(1)
8
9
10 noodles.moveRight(1)
11 snake()
12 noodles.jumpUp(1)
13 snake()
14 noodles.moveRight(1)
15 snake()
16 noodles.jumpUp(1)
```

#55. Bouncy Snake

Level Overview and Solutions

Bouncy Snake

Intro

Finally! Noodles found a glow fruit! She just needs a little more help getting there.

Default Code

```
1 def jumpSnake():
2     noodles.jumpUp(1)
3     noodles.moveRight(1)
4     noodles.jumpDown(1)
5     noodles.moveRight(1)
6     # add an action here to finish the function
```

Bouncy Snake Solution

```
1 def jumpSnake():
2     noodles.jumpUp(1)
3     noodles.moveRight(1)
4     noodles.jumpDown(1)
5     noodles.moveRight(1)
6     # add an action here to finish the function
7     noodles.jumpUp(1)
8
9
10 noodles.moveRight()
11 jumpSnake()
12 noodles.jumpRight(1)
13 noodles.jumpDown(1)
14 noodles.moveRight(1)
15 jumpSnake()
16 noodles.moveUp(1)
17 noodles.moveUp(2)
18 jumpSnake()
```

#60. Elevator

Level Overview and Solutions

Elevator

Intro

More glow fruits means we're getting closer to building the Acodus. I knew my Noodles could do it.

Default Code

```
1 def deliver(steps):
2     noodles.moveUp(steps)
3     noodles.take()
4     noodles.moveDown(steps)
5     noodles.drop()
6
7 noodles.moveRight(1)
8 deliver(3)
9 noodles.jumpRight(1)
10 deliver(4)
11 noodles.jumpRight(1)
12 deliver(2)
```

Elevator Solution

```
1 def deliver(steps):
2     noodles.moveUp(steps)
3     noodles.take()
4     noodles.moveDown(steps)
5     noodles.drop()
6
7 noodles.moveRight(1)
8 deliver(3)
9 noodles.jumpRight(1)
10 deliver(4)
11 noodles.jumpRight(1)
12 deliver(2)
13 noodles.moveRight(1)
14 noodles.moveUp(1)
15 noodles.jumpUp(1)
16 noodles.moveUp(1)
```

#63. Conveyor Belt

Level Overview and Solutions

Conveyor Belt

Intro

The glow fruits are more scattered here. Noodles will need help delivering them to the portals.

Default Code

```
1 def deliver(steps):
2     noodles.moveRight(steps)
3     noodles.take()
4     noodles.moveLeft(steps)
5     #Add a line to help Noodles drop the fruit
6
7
8
9     noodles.moveUp(1)
10 deliver(2)
11 noodles.jumpUp(1)
```

Conveyor Belt Solution

```
1 def deliver(steps):
2     noodles.moveRight(steps)
3     noodles.take()
4     noodles.moveLeft(steps)
5     noodles.drop()
6
7
8     noodles.moveUp(1)
9     deliver(2)
10    noodles.jumpUp(1)
11    deliver(4)
12    noodles.jumpUp(1)
13    deliver(3)
14    noodles.moveRight(3)
15    noodles.jumpRight(1)
```


#64. Corners

Level Overview and Solutions

Corners

Intro

Noodles has found more glow fruits! She's a clever little one. Let's help her deliver them, shall we?

Default Code

```
1 def deliver(upDownSteps, rightLeftsteps):
2     noodles.moveUp(upDownSteps)
3     noodles.moveRight(rightLeftsteps)
4     noodles.take()
5     noodles.moveLeft(rightLeftsteps)
6     noodles.moveDown(upDownSteps)
7     noodles.drop()
8
9
10 deliver(2, 4)
```

Corners Solution

```
1 def deliver(upDownSteps, rightLeftsteps):
2     noodles.moveUp(upDownSteps)
3     noodles.moveRight(rightLeftsteps)
4     noodles.take()
5     noodles.moveLeft(rightLeftsteps)
6     noodles.moveDown(upDownSteps)
7     noodles.drop()
8
9 deliver(2, 4)
10 deliver(3, 3)
11 deliver(4, 4)
12 deliver(5, 2)
13 noodles.moveRight(4)
14 noodles.jumpRight(1)
```

#65. Argumented Rectangles

Level Overview and Solutions

Argumented Rectangles

Intro

My poor Noodles looks tired. Let's finish gathering these glow fruits. Then she can rest for a bit.

Default Code

```
1 def deliver(jumpNumber, upDownSteps):
2     noodles.jumpRight(jumpNumber)
3     noodles.take()
4     noodles.moveUp(upDownSteps)
5     noodles.moveLeft(jumpNumber * 2)
6     # add an action here to finish the function
```

Argumented Rectangles Solution

```
1 def deliver(jumpNumber, upDownSteps):
2     noodles.jumpRight(jumpNumber)
3     noodles.take()
4     noodles.moveUp(upDownSteps)
5     noodles.moveLeft(jumpNumber * 2)
6     noodles.moveDown(upDownSteps)
7     noodles.drop()
8
9 deliver(1, 2)
10 deliver(2, 4)
11 deliver(3, 5)
12 noodles.jumpRight(3)
13 noodles.moveRight(1)
14
15 noodles.moveUp(5)
```

#69. Steady Delivery

Level Overview and Solutions

Steady Delivery

Intro

We don't have much time so we need to help Noodles collect the glow fruits as quickly as possible.

Default Code

```
1 def deliver(fruits):
2     noodles.moveRight(4)
3     noodles.take(fruits)
4     noodles.moveLeft(4)
5     noodles.drop()
6
7 fruitNumber = 7 # CHANGE THIS NUMBER
8
9 while True:
10     deliver(fruitNumber)
```

Steady Delivery Solution

```
1 def deliver(fruits):
2     noodles.moveRight(4)
3     noodles.take(fruits)
4     noodles.moveLeft(4)
5     noodles.drop()
6
7 fruitNumber = 4
8
9 while True:
10     deliver(fruitNumber)
```

#70. Light Delivery

Level Overview and Solutions

Light Delivery

Intro

Noodles has found a different kind of glow fruit. They're a lot lighter so she should be able to deliver more this time!

Default Code

```
1 def deliver(fruits, deliverSpeed):
2     noodles.setSpeed(deliverSpeed)
3     noodles.moveRight(5)
4     noodles.take(fruits)
5     noodles.moveLeft(5)
6     noodles.drop()
7
8
9 fruitNumber = 3 # CHANGE THIS NUMBER
10 speed = 15 # CHANGE THIS NUMBER
11
12 while True:
13     deliver(fruitNumber, speed)
```

Light Delivery Solution

```
1 def deliver(fruits, deliverSpeed):
2     noodles.setSpeed(deliverSpeed)
3     noodles.moveRight(5)
4     noodles.take(fruits)
5     noodles.moveLeft(5)
6     noodles.drop()
7
8
9 fruitNumber = 7
10 speed = 20
11
12 while True:
13     deliver(fruitNumber, speed)
```


#71. Heavy Delivery

Level Overview and Solutions

Heavy Delivery

Intro

There's not much time left and we still need more glow fruit. This fruit is heavier so you'll need to use **trial and error** again.

Default Code

```
1 def deliver(fruits, deliverySpeed):
2     noodles.setSpeed(deliverySpeed)
3     noodles.moveRight(5)
4     noodles.take(fruits)
5     noodles.moveLeft(5)
6     noodles.drop()
7
8
9 fruitNumber = 1 # CHANGE THIS NUMBER
10 speed = 50 # CHANGE THIS NUMBER
11
12 while True:
13     deliver(fruitNumber, speed)
```

Heavy Delivery Solution

```
1 def deliver(fruits, deliverySpeed):
2     noodles.setSpeed(deliverySpeed)
3     noodles.moveRight(5)
4     noodles.take(fruits)
5     noodles.moveLeft(5)
6     noodles.drop()
7
8
9 fruitNumber = 4
10 speed = 20
11
12 while True:
13     deliver(fruitNumber, speed)
```


#73. Optional Delivery

Level Overview and Solutions

Optional Delivery

Intro

This is the last of the glow fruit that we need! Please be quick. I can only delay the Fox Prince for so long. . .

Default Code

```
1 def deliver(fruits, deliverySpeed, steps):
2     noodles.setSpeed(deliverySpeed)
3     noodles.moveRight(steps)
4     noodles.take(fruits)
5     noodles.moveLeft(steps)
6     noodles.drop()
7
8
9 fruitNumber = 1 # CHANGE THIS NUMBER
10 speed = 50 # CHANGE THIS NUMBER
11 stepsToFruits = 1 # CHANGE THIS NUMBER
12
13 while True:
14     deliver(fruitNumber, speed, stepsToFruits)
```

Optional Delivery Solution

```
1 def deliver(fruits, deliverySpeed, steps):
2     noodles.setSpeed(deliverySpeed)
3     noodles.moveRight(steps)
4     noodles.take(fruits)
5     noodles.moveLeft(steps)
6     noodles.drop()
7
8
9 fruitNumber = 4
10 speed = 30
11 stepsToFruits = 3
12
13 while True:
14     deliver(fruitNumber, speed, stepsToFruits)
```

#78. Functional Crossing

Level Overview and Solutions

Functional Crossing

Intro

So many bridges and so little time. Functions should help us get across these rivers quickly.

Default Code

```
1 def crossRiver():
2     # Add function code here
3
4 crossRiver()
5 hero.moveUp(2)
6 crossRiver()
7 crossRiver()
```

Functional Crossing Solution

```
1 def crossRiver():
2     hero.jumpRight()
3     mouse.moveRight(2)
4
5
6 crossRiver()
7 hero.moveUp(2)
8 crossRiver()
9 crossRiver()
10
11 mouse.digRight()
12 mouse.moveDown()
13 hero.moveRight(2)
14 mouse.moveDown(2)
```

#79. Open Sesame

Level Overview and Solutions

Open Sesame

Intro

The Phoenix Lands is a mess! You'll need Mouse's help to get through all the obstacles.

Default Code

```
1 def zigZag():
2     # Add function code here
3
4 mouse.moveRight()
5 mouse.moveDown()
6 zigZag()
7 mouse.moveUp()
8 mouse.moveRight(2)
9 zigZag()
10 mouse.moveRight(2)
```

Open Sesame Solution

```
1 def zigZag():
2     hero.moveUp()
3     hero.moveRight()
4     hero.moveDown()
5     hero.moveRight()
6     hero.moveUp()
7
8 mouse.moveRight()
9 mouse.moveDown()
10 zigZag()
11 mouse.moveUp()
12 mouse.moveRight(2)
13 zigZag()
14 mouse.moveRight(2)
15 zigZag()
16 mouse.moveRight(2)
17 hero.jumpUp()
```


#82. Knocking on Haven's Door

Level Overview and Solutions

Knocking on Haven's Door

Intro

They call this the "Land of Many Gates". Once upon a time it was a mighty fortress that separated Curon the dragon lands from the human kingdom. Now it's just ruins you need to cross.

Default Code

```
1 # Define your function here
2     # Add function code here
3
4 mouse.moveDown()
5 stairStep()
6 mouse.moveUp()
7 mouse.digRight()
8 mouse.moveRight()
9 stairStep()
10 mouse.moveDown()
```

Knocking on Haven's Door Solution

```
1 def stairStep():
2     hero.moveRight()
3     hero.moveDown()
4     hero.moveRight()
5
6 mouse.moveDown()
7 stairStep()
8 mouse.moveUp()
9 mouse.digRight()
10 mouse.moveRight()
11 stairStep()
12 mouse.moveDown()
13 stairStep()
14 mouse.moveRight()
15 mouse.digRight()
16 mouse.moveRight()
17 stairStep()
18 hero.moveUp(4)
19 mouse.moveUp(5)
```

#83. Bridges of Huaraz County

Level Overview and Solutions

Bridges of Huaraz County

Intro

Huaraz is just beyond these bridges. We'll find Salazar there... but wait, can it be? The Splinters have taken over this town as well!

Default Code

```
1 # Define and write the stairStep function here.  
2  
3 hero.jumpRight()  
4 hero.castEarthWave()  
5 hero.moveUp()  
6 stairStep()
```

Bridges of Huaraz County Solution

```
1 def stairStep():  
2     mouse.moveRight(3)  
3     mouse.digDown()  
4  
5 hero.jumpRight()  
6 hero.castEarthWave()  
7 hero.moveUp()  
8 stairStep()  
9 hero.moveRight()  
10 hero.moveUp()  
11 enemy = hero.findNearestEnemy()  
12 hero.castEarthPit(enemy)  
13 stairStep()  
14 hero.jumpRight()  
15 stairStep()  
16 hero.moveRight()  
17 hero.jumpRight()  
18 hero.moveRight()
```

#86. Functional Crossing

Level Overview and Solutions

Functional Crossing

Intro

The path to my vault is like a maze, so you'll need much more powerful functions. Do you remember **parameters**? I taught them to you long ago. . .

Default Code

```
1 def caterpillar(upDownSteps, rightSteps):  
2     hero.moveUp(upDownSteps)  
3  
4  
5  
6  
7  
8 caterpillar(1, 2)  
9 caterpillar(2, 1)  
10 caterpillar(3, 2)  
11  
12 mouse.moveRight(1)  
13 mouse.digRight()
```

Functional Crossing Solution

```
1 def caterpillar(upDownSteps, rightSteps):
2     hero.moveUp(upDownSteps)
3     hero.moveRight(rightSteps)
4     hero.moveDown(upDownSteps)
5     hero.moveRight(1)
6
7
8 caterpillar(1, 2)
9 caterpillar(2, 1)
10 caterpillar(3, 2)
11
12 mouse.moveRight(1)
13 mouse.digRight()
14 mouse.moveRight(1)
15 mouse.digRight()
16
17 mouse.bark()
18
19 hero.sneakUp(7)
20 mouse.moveUp(2)
```

#89. L Rivers

Level Overview and Solutions

L Rivers

Intro

I warned you, my friend. The Darkness increases as we get closer to the vault. Be careful and don't get caught!

Default Code

```
1 def crossRiver(heroSteps, mouseSteps):
2     # Add function code here.
3
4
5
6
7
8 crossRiver(1, 2)
9 crossRiver(1, 1)
```

L Rivers Solution

```
1 def crossRiver(heroSteps, mouseSteps):
2     hero.jumpRight()
3     hero.moveUp(heroSteps)
4     mouse.moveUp(mouseSteps)
5     mouse.moveRight(2)
6
7
8 crossRiver(1, 2)
9 crossRiver(1, 1)
10 crossRiver(2, 3)
11
12 mouse.digRight()
13 hero.sneakUp()
14 hero.sneakRight(2)
15 hero.moveUp(1)
```

#93. Through Walls

Level Overview and Solutions

Through Walls

Intro

Good old Snikrep came through! Now we can reverse some of the damage the Darkness caused by restoring the poor Tengshe.

Default Code

```
1 def crossWall(heroSteps, mouseSteps):  
2     # Add function code here.  
3  
4  
5  
6  
7  
8  
9 crossWall(1, 1)  
10 crossWall(2, 1)  
11 crossWall(1, 4)  
12  
13 mouse.bark()  
14 enemy = hero.findNearestEnemy()  
15 hero.castTimeRestore(enemy)
```

Through Walls Solution

```
1 def crossWall(heroSteps, mouseSteps):
2     mouse.digRight()
3     mouse.moveDown(mouseSteps)
4     hero.moveUp(heroSteps)
5     hero.moveRight(3)
6     mouse.moveRight()
7
8
9 crossWall(1, 1)
10 crossWall(2, 1)
11 crossWall(1, 4)
12
13 mouse.bark()
14 enemy = hero.findNearestEnemy()
15 hero.castTimeRestore(enemy)
16 hero.moveUp(3);
17 mouse.moveDown(1);|
```

#94. Back to the Good Times

Level Overview and Solutions

Back to the Good Times

Intro

Salazar says we're getting closer to the Code Totems. I'm a little concerned on where all these Tengshe are coming from? I think the Darkness might know what we're up to.

Default Code

```
1 def pushAndRestore(steps):  
2     # Add function code here.  
3  
4  
5  
6  
7  
8  
9 pushAndRestore(2)  
10 pushAndRestore(3)
```

Back to the Good Times Solution

```
1 def pushAndRestore(steps):
2     hero.moveRight(steps)
3     mouse.moveRight(steps)
4     mouse.bark()
5     enemy = hero.findNearestEnemy()
6     hero.castTimeRestore(enemy)
7
8
9 pushAndRestore(2)
10 pushAndRestore(3)
11 pushAndRestore(1)
12
13 mouse.digUp()
14 mouse.moveUp(1)
15 hero.moveRight(1)
16 hero.moveUp(5)
17 hero.moveRight(1)
18 mouse.moveRight(2)
```

#95. Vault Door

Level Overview and Solutions

Vault Door

Intro

We're so close! The **Code Totems** are just beyond that **gate**. But watch out for those Tengshe! Time to use the `restore` spell to get us through.

Default Code

```
1 # Define the function pushAndRestore with 2 parameters
2   # Add function code here.
3
4
5
6
7
8
9
10
11 pushAndRestore(2, 2)
```

Vault Door Solution

```
1 def pushAndRestore(downSteps, rightSteps):
2     hero.moveDown(downSteps)
3     hero.moveRight(rightSteps)
4     mouse.moveRight(rightSteps)
5     mouse.moveDown(downSteps)
6     mouse.bark()
7     enemy = hero.findNearestEnemy()
8     hero.castTimeRestore(enemy)
9
10
11 pushAndRestore(2, 2)
12 pushAndRestore(1, 1)
13 pushAndRestore(1, 2)
14
15 mouse.moveUp(1)
16 mouse.digUp()
17 mouse.moveRight(1)
18 hero.moveRight(2)
19 hero.jumpUp()
20 hero.jumpUp()
21 hero.jumpUp()
22 mouse.moveUp()
23 hero.moveUp(1)
```

#97. Refoxion

Level Overview and Solutions

Refoxion

Intro

Ugh. My head. Someone hit me from behind. Octans! He has the Code Totems! You must stop him with the `restore` spell!

Default Code

```
1 octans = 'Octans'
```

Refoxion Solution

```
1 octans = 'Octans'
2
3 def pushAndRestore(upSteps, rightSteps):
4     hero.moveRight(rightSteps)
5     hero.moveUp(upSteps)
6     mouse.moveUp(upSteps)
7     mouse.moveRight(rightSteps)
8     mouse.bark()
9     hero.castTimeRestore(octans)
10
11
12 pushAndRestore(2, 2)
13 pushAndRestore(1, 3)
14 pushAndRestore(1, 2)
15
16 hero.sneakRight(1)
17 hero.sneakDown(2)
18 hero.castTimeRestore(octans)
```

#105. Yum Yum Gems

Level Overview and Solutions

Yum Yum Gems

Intro

Ok, I think you'll like the game I made. It's full of gems and lots of eating.

Default Code

```

1 # GAME SET UP
2 # Spawns the objects for the game: the avatar, the gem, and the enemy.
3 avatar = game.spawnObject("turtle")
4 avatar.setControl("arrows")
5
6 gem = game.spawnObject("gem-green", "random", "random")
7
8 enemy = game.spawnObject("spider")
9 enemy.setChaseTarget(gem)
10 enemy.setSpeed(0.8)
11
12 # Shows the directions of the game to the player.
13 game.addTextDirections("Use arrow keys to help the turtle eat the gems before the enemy
    does!")
14
15 points = 0
16
17 # Defines a function that moves the gem to a random location.
18 def moveGem():
19     gem.setX("random")
20     gem.setY("random")
21
22 # GAME LOOP
23 # The while loop repeats rounds of the game until 60 seconds have passed.
24 # Each round updates displays and checks if objects have hit the gem.
25 while game.timePassed() < 60:
26     game.advanceGame()
27     game.setDisplay("TIME", game.timePassed())
28     game.setDisplay("SCORE", points)
29     if avatar.hit(gem):
30         points += 1
31         moveGem()
32     if enemy.hit(gem):
33         moveGem()
34
35
36
37 # GAME END
38 # After the loop ends, the conditional checks the point score.
39 # Depending on the number of points, the player sees a win or an end screen.
40 if points >= 10:
41     game.win("You got " + points + " points! You Win!")
42 else:
43     game.end("You got <20 points! Try Again.")

```

Yum Yum Gems Solution

```
1 # GAME SET UP
2 # Spawns the objects for the game: the avatar, the gem, and the enemy.
3 avatar = game.spawnObject("turtle")
4 avatar.setControl("arrows")
5
6 gem = game.spawnObject("gem-green", "random", "random")
7
8 enemy = game.spawnObject("spider")
9 enemy.setChaseTarget(gem)
10 enemy.setSpeed(0.8)
11
12 # Shows the directions of the game to the player.
13 game.addTextDirections("Use arrow keys to help the turtle eat the gems before the enemy
    does!")
14
15 points = 0
16
17 # Defines a function that moves the gem to a random location.
18 def moveGem():
19     gem.setX("random")
20     gem.setY("random")
21
22 # GAME LOOP
23 # The while loop repeats rounds of the game until 60 seconds have passed.
24 # Each round updates displays and checks if objects have hit the gem.
25 while game.timePassed() < 60:
26     game.advanceGame()
27     game.setDisplay("TIME", game.timePassed())
28     game.setDisplay("SCORE", points)
29     if avatar.hit(gem):
30         points += 1
31         moveGem()
32     if enemy.hit(gem):
33         moveGem()
34
35
36
37 # GAME END
38 # After the loop ends, the conditional checks the point score.
39 # Depending on the number of points, the player sees a win or an end screen.
40 if points >= 10:
41     game.win("You got " + points + " points! You Win!")
42 else:
43     game.end("You got <20 points! Try Again.")
```

#110. Rock Catches Leaf

Level Overview and Solutions

Rock Catches Leaf

Intro

Don't you love the wonderful smell of **leaves** in the fall? My game starts with leaves placed perfectly in a special **message**.

Default Code

```

1 # GAME SET UP
2 # Creates the rock and displays the number of rocks that players get.
3 rock = game.spawnObject("rock-clay", 4.5, 2)
4
5 # Creates a paddle made of 3 boxes that is moved together with arrow keys.
6 paddle1 = game.spawnObject("box", 4, 0)
7 paddle2 = game.spawnObject("box", 4.5, 0)
8 paddle3 = game.spawnObject("box", 5, 0)
9 paddle1.setControl("arrows")
10 paddle2.setControl("arrows")
11 paddle3.setControl("arrows")
12
13 # Creates leaves that spell out the word "OK" for the rock to hit.
14 game.spawnObject("leaf-red", 1, 6)
15 game.spawnObject("leaf-red", 1, 5)
16 game.spawnObject("leaf-red", 1, 4)
17 game.spawnObject("leaf-red", 1, 3)
18 game.spawnObject("leaf-red", 2, 6)
19 game.spawnObject("leaf-red", 2, 3)
20 game.spawnObject("leaf-red", 3, 6)
21 game.spawnObject("leaf-red", 3, 5)
22 game.spawnObject("leaf-red", 3, 4)
23 game.spawnObject("leaf-red", 3, 3)
24 game.spawnObject("leaf-red", 5, 6)
25 game.spawnObject("leaf-red", 5, 5)
26 game.spawnObject("leaf-red", 5, 4)
27 game.spawnObject("leaf-red", 5, 3)
28 game.spawnObject("leaf-red", 6, 5)
29 game.spawnObject("leaf-red", 6, 4)
30 game.spawnObject("leaf-red", 7, 6)
31 game.spawnObject("leaf-red", 7, 3)
32 game.spawnObject("leaf-red", 9, 3)
33
34 # Shows the directions of the game.
35 game.addTextDirections("Catch all the leaves!")
36 game.setBorders("infinite")
37
38 #Creates global variables that will be used during the game.
39 leavesToCatch = 19 #the number of leaves spawned above
40 numOfRocks = 3
41 ROCK_SPEED = 0.1
42 rockVX = -ROCK_SPEED
43 rockVY = ROCK_SPEED
44
45 # Displays the number of rocks to the player.
46 game.setDisplay("rocks", numOfRocks)
47
48 # This function resets the rock to its starting location and movement direction.
49 def initRock():
50     global rockVX, rockVY, ROCK_SPEED
51     rockVX = -ROCK_SPEED
52     rockVY = ROCK_SPEED
53     rock.setXY(4.5, 1.5)
54
55 # These functions set up how the rock moves when it hits an object.
56 def bounceLeftRight():
57     global rockVX

```

```

57     global rockVX
58     rockVX = -rockVX
59
60 def bounceUpDown():
61     global rockVY
62     rockVY = -rockVY
63
64 # This function is called if the rock hits a leaf.
65 def removeLeaf(leaf):
66     global leavesToCatch
67     # The rock's direction is changed.
68     if leaf.getY() - rock.getY() > .4:
69         bounceUpDown()
70     elif leaf.getY() - rock.getY() < -.4:
71         bounceUpDown()
72     else:
73         bounceLeftRight()
74     # The leaf that is caught is hidden.
75     leaf.hide()
76     leavesToCatch = leavesToCatch - 1
77
78     #If all the leaves are caught, then the player gets the win screen!
79     if leavesToCatch == 0:
80         rock.hide()
81         game.win("You Did It!")
82
83 # GAME LOOP
84 # The for loop plays 9999 rounds of the game.
85 # Look at comments inside the loop to find out what happens in each round.
86 for i in range(9999):
87     game.advanceGame()
88
89     # Moves the rock forward with its current direction and speed.
90     rock.setX(rock.getX() + rockVX)
91     rock.setY(rock.getY() + rockVY)
92
93     paddle1.setY(0) #only move left/right
94     paddle2.setY(0) #only move left/right
95     paddle3.setY(0) #only move left/right
96
97     # Checks to see if the rock hits the paddle.
98     # If so, it changes the rock's direction, depending on its location.
99     hitPaddle = rock.hit("box")
100    if hitPaddle:
101        rockVY = ROCK_SPEED
102    # Checks to see if the rock hits a side of the game screen.
103    # If so, it changes the rock's direction, depending on its location.
104    if rock.getX() <= -0.2:
105        rockVX = ROCK_SPEED
106    if rock.getX() > 9.2:
107        rockVX = -ROCK_SPEED
108    if rock.getY() > 7.2:
109        rockVY = -ROCK_SPEED
110
111    # Checks to see if the rock hits any of the leaves.
112    #If so, it calls the removeLeaf function using the leaf that's hit.
113    rockHitLeaf = rock.hit("leaf-red")

```

```
114     if rockHitLeaf:
115         removeLeaf(rock.findNearest("leaf-red"))
116
117     # Checks to see if the rock has fallen to the bottom of the screen.
118     # If so, then the players loses a rock.
119     if rock.getY() < -0.5:
120         numOfRocks = numOfRocks - 1
121         game.setDisplay("rocks", numOfRocks)
122         if numOfRocks > 0:
123             initRock()
124         else:
125             rock.hide()
126             game.end("WOMP WOMP")
```

Rock Catches Leaf Solution


```

1 # GAME SET UP
2 # Creates the rock and displays the number of rocks that players get.
3 rock = game.spawnObject("rock-clay", 4.5, 2)
4
5 # Creates a paddle made of 3 boxes that is moved together with arrow keys.
6 paddle1 = game.spawnObject("box", 4, 0)
7 paddle2 = game.spawnObject("box", 4.5, 0)
8 paddle3 = game.spawnObject("box", 5, 0)
9 paddle1.setControl("arrows")
10 paddle2.setControl("arrows")
11 paddle3.setControl("arrows")
12
13 # Creates leaves that spell out the word "OK" for the rock to hit.
14 game.spawnObject("leaf-red", 1, 6)
15 game.spawnObject("leaf-red", 1, 5)
16 game.spawnObject("leaf-red", 1, 4)
17 game.spawnObject("leaf-red", 1, 3)
18 game.spawnObject("leaf-red", 2, 6)
19 game.spawnObject("leaf-red", 2, 3)
20 game.spawnObject("leaf-red", 3, 6)
21 game.spawnObject("leaf-red", 3, 5)
22 game.spawnObject("leaf-red", 3, 4)
23 game.spawnObject("leaf-red", 3, 3)
24 game.spawnObject("leaf-red", 5, 6)
25 game.spawnObject("leaf-red", 5, 5)
26 game.spawnObject("leaf-red", 5, 4)
27 game.spawnObject("leaf-red", 5, 3)
28 game.spawnObject("leaf-red", 6, 5)
29 game.spawnObject("leaf-red", 6, 4)
30 game.spawnObject("leaf-red", 7, 6)
31 game.spawnObject("leaf-red", 7, 3)
32 game.spawnObject("leaf-red", 9, 3)
33
34 # Shows the directions of the game.
35 game.addTextDirections("Catch all the leaves!")
36 game.setBorders("infinite")
37
38 #Creates global variables that will be used during the game.
39 leavesToCatch = 19 #the number of leaves spawned above
40 numOfRocks = 3
41 ROCK_SPEED = 0.1
42 rockVX = -ROCK_SPEED
43 rockVY = ROCK_SPEED
44
45 # Displays the number of rocks to the player.
46 game.setDisplay("rocks", numOfRocks)
47
48 # This function resets the rock to its starting location and movement direction.
49 def initRock():
50     global rockVX, rockVY, ROCK_SPEED
51     rockVX = -ROCK_SPEED
52     rockVY = ROCK_SPEED
53     rock.setXY(4.5, 1.5)
54
55 # These functions set up how the rock moves when it hits an object.
56 def bounceLeftRight():
57     global rockVX

```

```

57     global rockVX
58     rockVX = -rockVX
59
60 def bounceUpDown():
61     global rockVY
62     rockVY = -rockVY
63
64 # This function is called if the rock hits a leaf.
65 def removeLeaf(leaf):
66     global leavesToCatch
67     # The rock's direction is changed.
68     if leaf.getY() - rock.getY() > .4:
69         bounceUpDown()
70     elif leaf.getY() - rock.getY() < -.4:
71         bounceUpDown()
72     else:
73         bounceLeftRight()
74     # The leaf that is caught is hidden.
75     leaf.hide()
76     leavesToCatch = leavesToCatch - 1
77
78     #If all the leaves are caught, then the player gets the win screen!
79     if leavesToCatch == 0:
80         rock.hide()
81         game.win("You Did It!")
82
83 # GAME LOOP
84 # The for loop plays 9999 rounds of the game.
85 # Look at comments inside the loop to find out what happens in each round.
86 for i in range(9999):
87     game.advanceGame()
88
89     # Moves the rock forward with its current direction and speed.
90     rock.setX(rock.getX() + rockVX)
91     rock.setY(rock.getY() + rockVY)
92
93     paddle1.setY(0) #only move left/right
94     paddle2.setY(0) #only move left/right
95     paddle3.setY(0) #only move left/right
96
97     # Checks to see if the rock hits the paddle.
98     # If so, it changes the rock's direction, depending on its location.
99     hitPaddle = rock.hit("box")
100    if hitPaddle:
101        rockVY = ROCK_SPEED
102    # Checks to see if the rock hits a side of the game screen.
103    # If so, it changes the rock's direction, depending on its location.
104    if rock.getX() <= -0.2:
105        rockVX = ROCK_SPEED
106    if rock.getX() > 9.2:
107        rockVX = -ROCK_SPEED
108    if rock.getY() > 7.2:
109        rockVY = -ROCK_SPEED
110
111    # Checks to see if the rock hits any of the leaves.
112    #If so, it calls the removeLeaf function using the leaf that's hit.
113    rockHitLeaf = rock.hit("leaf-red")

```

```
114     if rockHitLeaf:
115         removeLeaf(rock.findNearest("leaf-red"))
116
117     # Checks to see if the rock has fallen to the bottom of the screen.
118     # If so, then the players loses a rock.
119     if rock.getY() < -0.5:
120         numOfRocks = numOfRocks - 1
121         game.setDisplay("rocks", numOfRocks)
122         if numOfRocks > 0:
123             initRock()
124         else:
125             rock.hide()
126             game.end("WOMP WOMP")
```

#112. Moth Battle

Level Overview and Solutions

Moth Battle

Intro

My game is a mighty battle between a humble moth and endless time storms.

Default Code

```

1 # GAME SET UP
2
3 # Creates the objects for the game: a moth, a skull, and enemies.
4 enemy1 = game.spawnObject("storm","center", -1)
5 enemy2 = game.spawnObject("storm", "center", -1)
6 enemy3 = game.spawnObject("storm", "center", -1)
7
8 skull = game.spawnObject("skull","center", "bottom")
9 skull.setScale(0.5)
10
11 moth = game.spawnObject("moth","center", "bottom")
12 moth.setControl("wasd")
13
14 # Shows the directions to the player and sets up the border.
15 game.addTextDirections("Defeat the enemies with skulls of fire! Use WASD keys and don't
    get hit or touch the walls.")
16 game.setBorders("infinite")
17
18 # Creates variables that will be used during the game.
19 width = 6
20 mothSpeed = 0.1
21 skullSpeed = 0.2
22
23 points = 0
24 lives = 3
25
26 # Displays the points and lives for the player
27 game.setDisplay("points", points)
28 game.setDisplay("life", lives)
29
30 # These functions reset objects to new starting locations.
31 def initAvatar():
32     moth.setXY("center", "bottom")
33
34 def initEnemy(enemy):
35     mothSpeed = mothSpeed + 0.005
36     enemy.setXY(game.random(1, 8), 7)
37
38 def initSkull():
39     skull.setXY(moth.getX(), moth.getY())
40
41 # This function moves an enemy one step down the screen.
42 def enemyStep(enemy):
43     enemy.setY(enemy.getY() - mothSpeed)
44
45 # This function checks if an enemy has reached the bottom of the game screen. If it has,
    then its location resets.
46 def checkEnemy(enemy):
47     if enemy.getY() > 0:
48         enemyStep(enemy)
49     else:
50         initEnemy(enemy)
51
52 # Builds a wall on the left and right side of the game screen.
53 for i in range(8):
54     game.spawnObject("wall", 0, i)
55     game.spawnObject("wall", 8, i)

```

```

55     game.spawnObject( wall , 5, 1)
56
57 # GAME LOOP
58 # The for loop plays 9999 rounds of the game.
59 # Look at comments inside the loop to find out what happens in each round.
60 for frame in range(9999):
61     game.advanceGame()
62     game.setDisplay("points", points)
63
64     # Checks the location of each enemy and moves them one step down the screen.
65     checkEnemy(enemy1)
66     checkEnemy(enemy2)
67     checkEnemy(enemy3)
68
69     # Moves the skull one step up the game screen.
70     skull.setY(skull.getY() + skullSpeed)
71
72     # Checks if the skull hits any enemy.
73     # If so, the player gets a point, the enemy resets, and the moth starts moving faster
74     skullHit = skull.hit("storm")
75     if skullHit:
76         points = points + 1
77         thisStorm = skull.findNearest("storm")
78         initEnemy(thisStorm) # respawn immediately
79         initSkull()
80         mothSpeed = mothSpeed + 0.01 # every time you defeat an enemy, the moth gets
            faster
81
82     # Moves the skull back to the moth once it moves off the screen.
83     if skull.getY() > 7.5:
84         initSkull()
85
86     # Checks if the moth has hit one of the enemies.
87     # If so, the player loses a life and the moth location is reset.
88     crashEnemy = moth.hit("storm")
89     if crashEnemy:
90         points = points + 1
91         lives = lives - 1
92         game.setDisplay("life", lives)
93         initAvatar()
94         # If the player runs out of lives, then the game ends.
95         if lives<1:
96             game.end("GAME OVER")
97
98     # Checks if the moth has hit one of the walls.
99     # If so, the game ends.
100    crashWall = moth.hit("wall")
101    if crashWall:
102        game.end("GAME OVER")
103
104    # Keeps the moth from moving off the screen
105    if moth.getY()>7:
106        moth.setY(7)
107    elif moth.getY()<0:
108        moth.setY(0)

```

Moth Battle Solution

```

1 # GAME SET UP
2
3 # Creates the objects for the game: a moth, a skull, and enemies.
4 enemy1 = game.spawnObject("storm","center", -1)
5 enemy2 = game.spawnObject("storm", "center", -1)
6 enemy3 = game.spawnObject("storm", "center", -1)
7
8 skull = game.spawnObject("skull","center", "bottom")
9 skull.setScale(0.5)
10
11 moth = game.spawnObject("moth","center", "bottom")
12 moth.setControl("wasd")
13
14 # Shows the directions to the player and sets up the border.
15 game.addTextDirections("Defeat the enemies with skulls of fire! Use WASD keys and don't
    get hit or touch the walls.")
16 game.setBorders("infinite")
17
18 # Creates variables that will be used during the game.
19 width = 6
20 mothSpeed = 0.1
21 skullSpeed = 0.2
22
23 points = 0
24 lives = 3
25
26 # Displays the points and lives for the player
27 game.setDisplay("points", points)
28 game.setDisplay("life", lives)
29
30 # These functions reset objects to new starting locations.
31 def initAvatar():
32     moth.setXY("center", "bottom")
33
34 def initEnemy(enemy):
35     mothSpeed = mothSpeed + 0.005
36     enemy.setXY(game.random(1, 8), 7)
37
38 def initSkull():
39     skull.setXY(moth.getX(), moth.getY())
40
41 # This function moves an enemy one step down the screen.
42 def enemyStep(enemy):
43     enemy.setY(enemy.getY() - mothSpeed)
44
45 # This function checks if an enemy has reached the bottom of the game screen. If it has,
    then its location resets.
46 def checkEnemy(enemy):
47     if enemy.getY() > 0:
48         enemyStep(enemy)
49     else:
50         initEnemy(enemy)
51
52 # Builds a wall on the left and right side of the game screen.
53 for i in range(8):
54     game.spawnObject("wall", 0, i)
55     game.spawnObject("wall", 8, i)

```



```

55     game.spawnObject( wall , 5, 1)
56
57 # GAME LOOP
58 # The for loop plays 9999 rounds of the game.
59 # Look at comments inside the loop to find out what happens in each round.
60 for frame in range(9999):
61     game.advanceGame()
62     game.setDisplay("points", points)
63
64     # Checks the location of each enemy and moves them one step down the screen.
65     checkEnemy(enemy1)
66     checkEnemy(enemy2)
67     checkEnemy(enemy3)
68
69     # Moves the skull one step up the game screen.
70     skull.setY(skull.getY() + skullSpeed)
71
72     # Checks if the skull hits any enemy.
73     # If so, the player gets a point, the enemy resets, and the moth starts moving faster
74     skullHit = skull.hit("storm")
75     if skullHit:
76         points = points + 1
77         thisStorm = skull.findNearest("storm")
78         initEnemy(thisStorm) # respawn immediately
79         initSkull()
80         mothSpeed = mothSpeed + 0.01 # every time you defeat an enemy, the moth gets
            faster
81
82     # Moves the skull back to the moth once it moves off the screen.
83     if skull.getY() > 7.5:
84         initSkull()
85
86     # Checks if the moth has hit one of the enemies.
87     # If so, the player loses a life and the moth location is reset.
88     crashEnemy = moth.hit("storm")
89     if crashEnemy:
90         points = points + 1
91         lives = lives - 1
92         game.setDisplay("life", lives)
93         initAvatar()
94         # If the player runs out of lives, then the game ends.
95         if lives<1:
96             game.end("GAME OVER")
97
98     # Checks if the moth has hit one of the walls.
99     # If so, the game ends.
100    crashWall = moth.hit("wall")
101    if crashWall:
102        game.end("GAME OVER")
103
104    # Keeps the moth from moving off the screen
105    if moth.getY()>7:
106        moth.setY(7)
107    elif moth.getY()<0:
108        moth.setY(0)

```


#115. Curiosity Sandbox

Level Overview and Solutions

Use functions, loops, and conditionals to make a custom game where objects interact with each other.

Intro

This is it: Time to defeat the Darkness once and for all. We only need your game to do it. Don't worry, my friend. I know you are ready.

Default Code

```
1 # GAME SET UP
2 # Spawn your avatar and set the control here.
3
4 # Spawn any additional objects you'll need for your game.
5
6 # Write the directions for your game
7
8
9 # Set up any object interactions here such as chasing or escaping.
10
11 # Set up any variables you'll need for your game such as "points" or "lives".
12 points = 0
13
14 # Use `setDisplay` to display information to the player.
15
16 # Define any functions here for code that you may want to reuse.
17
18 # GAME LOOP
19 while game.timePassed() < 60:
20     game.advanceGame() # Do not remove this method or it will break the game!
21     # Update your displays here
22     # Use conditionals to check whether objects have hit each other.
23
24 # GAME WIN or LOSE
25 # Add a conditional and use `game.win` and `game.end` to decide how the player wins or
    loses the game.
```

Curiosity Sandbox Solution

```

1 # GAME SET UP
2 # Spawn your avatar and set the control here.
3 # I got the chase game idea from Vega's code.
4 avatar = game.spawnObject("ghost",4,5)
5 avatar.setControl("wasd")
6 avatar.setSpeed(1.5)
7
8 # Spawn any additional objects you'll need for your game.
9 jewel1 = game.spawnObject("jewel-blue", "random", "random")
10 jewel2 = game.spawnObject("jewel-blue", "random", "random")
11 jewel3 = game.spawnObject("jewel-blue", "random", "random")
12
13 enemy = game.spawnObject("seeker", "center", "top")
14
15 # Write the directions for your game
16 game.addTextDirections("Use WASD keys to help the ghost eat the jewels before the enemy
    does!")
17
18 # Set up any object interactions here such as chasing or escaping.
19 enemy.setChaseTarget("jewel-blue")
20 enemy.setSpeed(1)
21
22 # Set up any variables you'll need for your game such as "points" or "lives".
23 points = 0
24
25 # Use `setDisplay` to display information to the player.
26 game.setDisplay("TIME", 0)
27 game.setDisplay("SCORE", points)
28
29 # Define any functions here for code that you may want to reuse.
30 def moveJewel(jewel):
31     jewel.setX("random")
32     jewel.setY("random")
33
34 # GAME LOOP
35 # The while loop repeats rounds of the game until 45 seconds have passed.
36 while game.timePassed() < 45:
37     game.advanceGame() # Do not remove this method or it will break the game!
38     # Update your displays here
39     game.setDisplay("TIME", game.timePassed())
40     game.setDisplay("SCORE", points)
41     # Use conditionals to check whether objects have hit each other.
42     # I got the idea of looking for a type of object from Capella's code.
43     if avatar.hit("jewel-blue"):
44         points += 1
45         moveJewel(avatar.findNearest("jewel-blue"))
46     if enemy.hit("jewel-blue"):
47         moveJewel(enemy.findNearest("jewel-blue"))
48
49 # GAME WIN or LOSE
50 # Add a conditional and use `game.win` and `game.end` to decide how the player wins or
    loses the game.
51 if points >= 15:
52     game.win("You got " + points + " points! You Win!")
53 else:
54     game.end("You got <15 points! Try again.")

```

